Experimental Determination of Moment of Inertia

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Contents

1 Abstract

1.1

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

2.1

Moment of inertia is a physical property of anything which rotates, charachterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate anc measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of mesuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

3.1

To calculate with rotational movement and can not idealize the mass to a particle as often done in translational calculations, you have to use expresions giving a link between the mass of an object and how it is distributed

$I = moment\ of\ inertia$ differen geometric forms will give diferent calculations about its centroid being the center of the distributed mass.

Examples of equations of moment of inertia
Equation for a circular plate

$$I_x = I_y = \frac{1}{2}I_z = m\frac{r^2}{4} \tag{1}$$

[1, p. 185]
Equation for a circular ring

$$I_x = I_y = \frac{1}{2}I_z = \frac{1}{2}mr^2 \tag{2}$$

[1, p. 185]

Using this link Newtons second law $\sum F = ma$ analogusly can be aplied for rotational movement.

$$\sum \tau = I\alpha \tag{3}$$

τ equals torsion I equals moment of inertia and α the angular acceleration

To calculate Torque (τ) it is the distance to where the aplied Force perpendikular to the rotational center see figure (fbd)

$$\tau = Tr \tag{4}$$

T is the aplied force in Newton and r is the distance to the point of perpendicular aplied force

Given an expresion for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha \tag{5}$$

In the case that the force is generated with a free faling weight the folowing substitutions can be made $F = mg = T$ giving the expression

$$mgr = I\alpha \tag{6}$$

When the centroid of an object and the axes of rotation dosent aligne The Paralell axel Theorem

$$I_{total} = I_{cm} + mr^2 \; (r = the\,distance\,between\,axis) \tag{7}$$

tourqe are given by T times the distance to the perpendicular line the force are aplied on

$$Tr = \tau \tag{8}$$

$$I = \frac{mgr}{\alpha} + mr^2 \tag{9}$$

4 Method

4.1

$$\sum F = mg - T = ma \tag{10}$$

a = the linear acceleration of the falling mass

5 Results

5.1

6 Discussion

6.1

7 References

7.1

References

[1] Carl Nordling and Jonny Österman. *Physics Handbook for Science and Engineering*. 9th ed. Studentlitteratur, 2020.

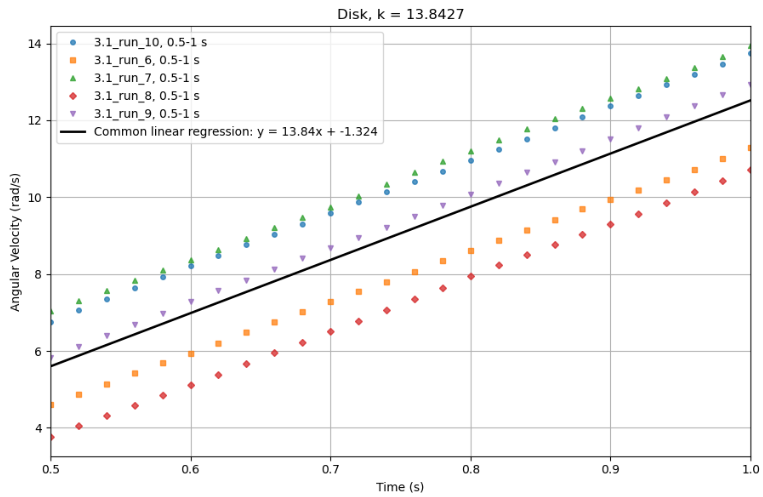

Figure 1: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

8 Appendix

8.1

9 Contributions

9.1

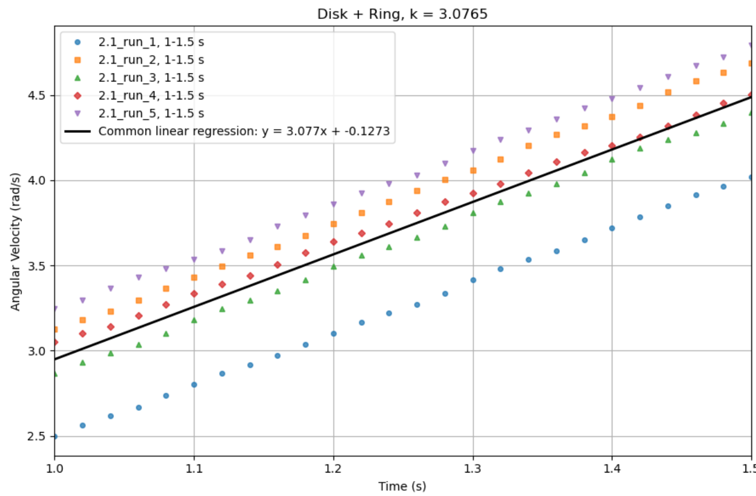

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

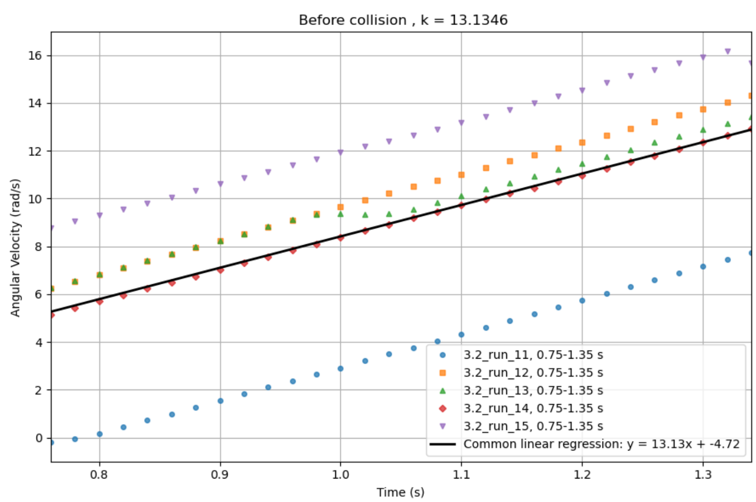

Figure 3: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

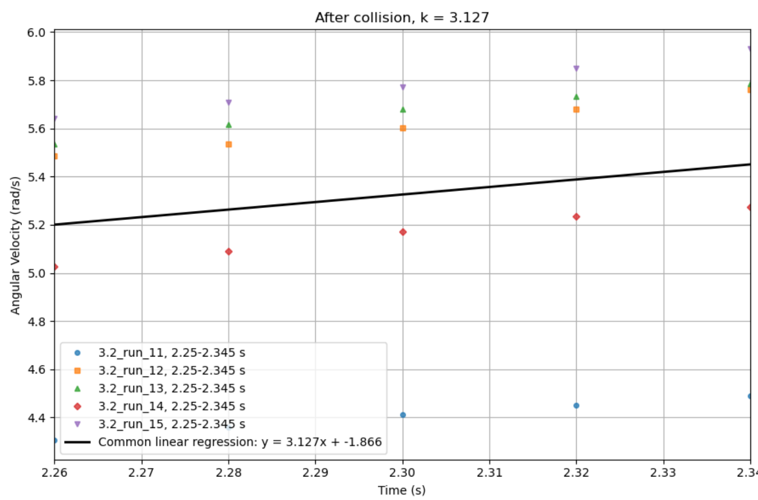

Figure 4: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.